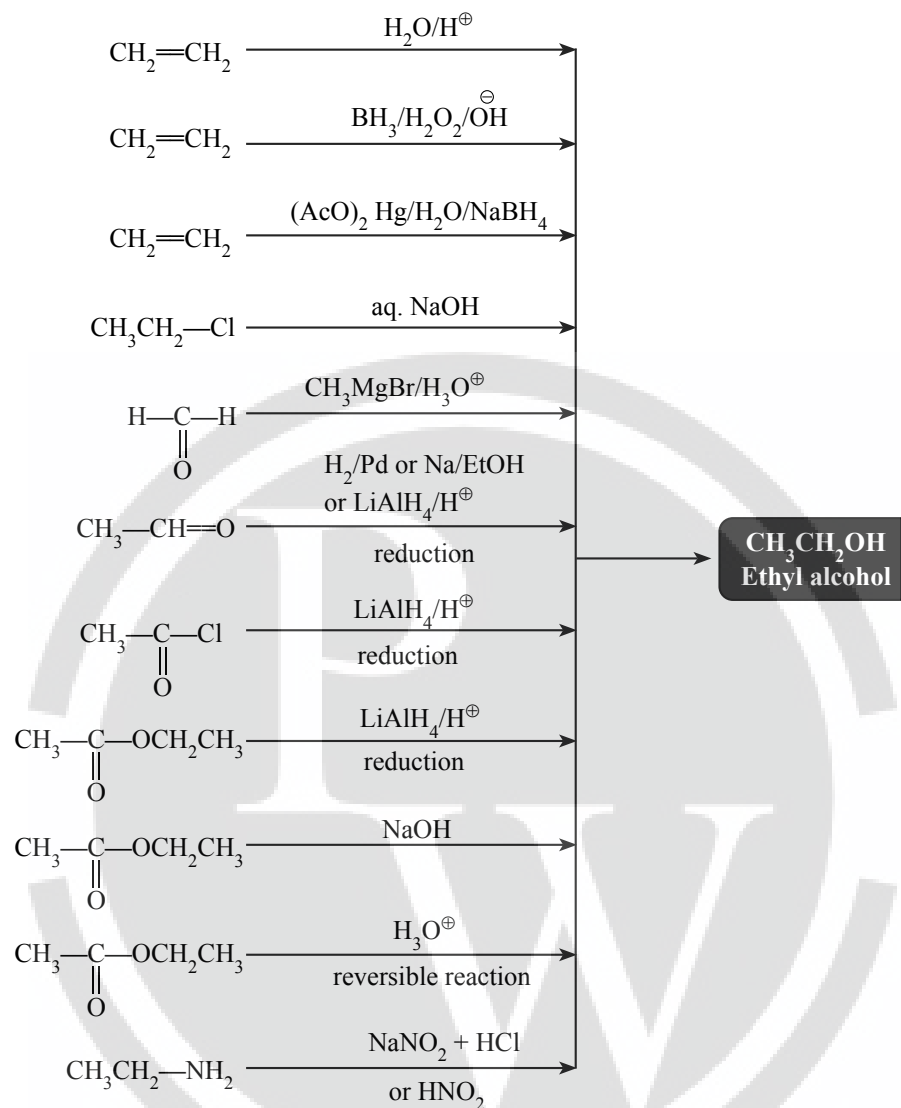


Alcohols, Phenols and Ethers

SHORT NOTES

METHODS OF PREPARATION OF ETHYL ALCOHOL



CHEMICAL REACTIONS OF ALCOHOL

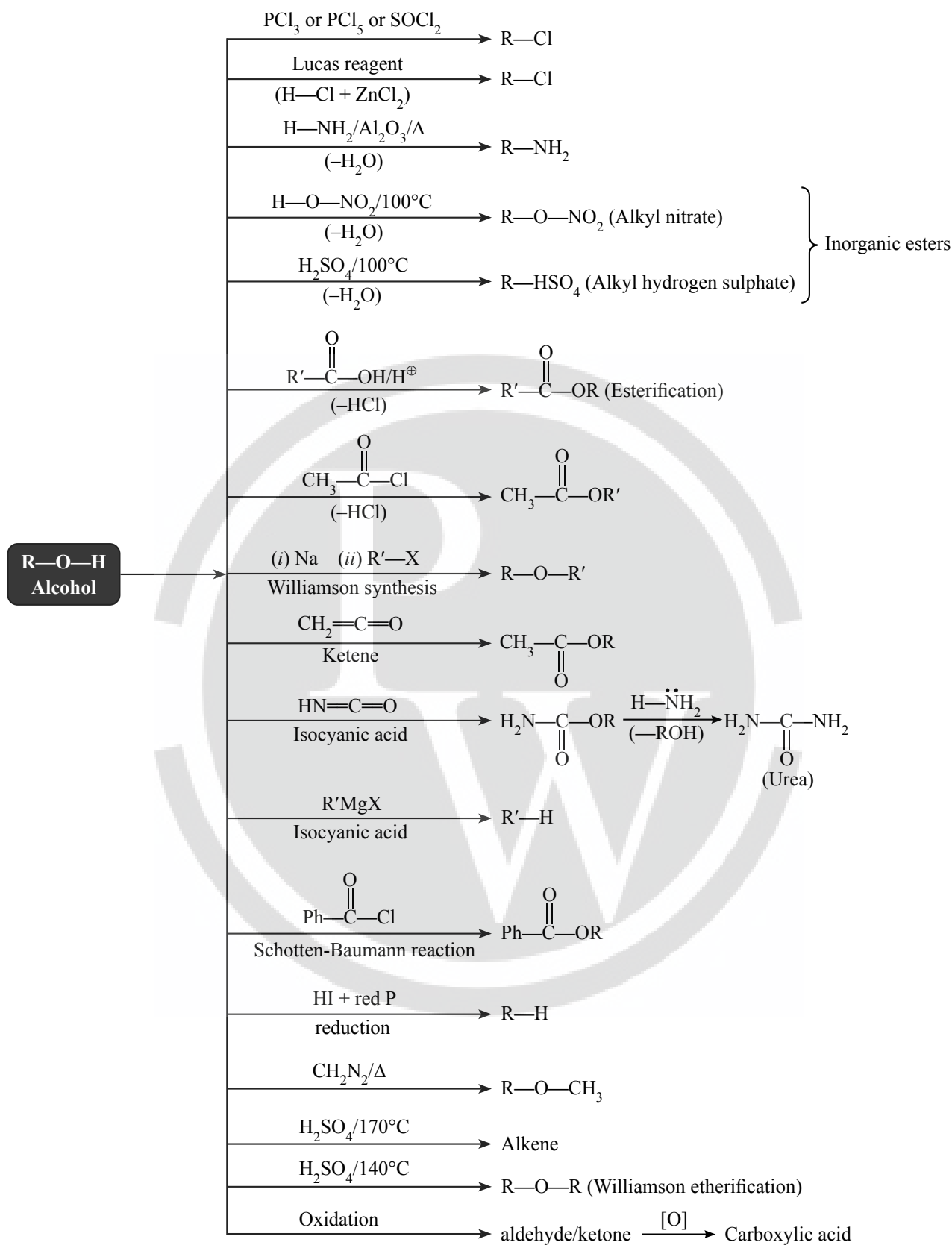


Table 10: Distinction of Primary, Secondary and Tertiary Alcohol

Distinguish between 1°, 2° and 3° Alcohol			
Reagent	$\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\overset{\text{OH}}{\underset{ }{\text{C}}}\text{H}_2$ 1° Alcohol	$\text{H}_3\text{C}-\text{CH}_2-\overset{\text{OH}}{\underset{ }{\text{CH}}}-\text{CH}_3$ 2° Alcohol	$\text{H}_3\text{C}-\overset{\text{CH}_3}{\underset{\text{CH}_3}{\underset{ }{\text{C}}}}-\text{OH}$ 3° Alcohol
PCC/PDC Anhy. CrO_3	$\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\overset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{H}$	$\text{H}_3\text{C}-\text{CH}_2-\overset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{CH}_3$	No reaction
$\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$ $\text{KMnO}_4/\text{H}^+/\text{OH}^-/\Delta$ Jones Reagent	$\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\overset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{OH}$	$\text{H}_3\text{C}-\overset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{OH} + \text{CH}_3-\overset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{OH}$	No reaction
$\text{Cu}/500^\circ\text{C}$	$\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\overset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{H}$	$\text{CH}_3-\text{CH}_2-\overset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{CH}_3$	$\text{H}_3\text{C}-\overset{\text{CH}_3}{\underset{\text{CH}_3}{\underset{ }{\text{C}}}}=\text{CH}_2$
Lucas Reagent HCl/ZnCl_2	Cloudiness appear upon heating after 30 mins.	Within five min.	Immediately

METHODS OF PREPARATION OF PHENOL

